

Lea Brody-Heine

lea_brody-heine@alumni.brown.edu | Software Engineer | **BA & MSc**
leabrodyheine.github.io | linkedin.com/in/lea-brody-heine | github.com/leabrodyheine

Software engineer with 2+ years of experience building production software, internal developer platforms, and AI-powered tooling. Strong written and verbal communication skills and proven ability to collaborate cross-functionally. Demonstrated ability to use AI to improve speed and quality in my day-to-day workflow for relevant outputs.

Education

University of St Andrews *MSc Computer Science*
GPA: First Class Honors with Distinction | Dean's List

St Andrews, Scotland
Graduated: 09/24

Brown University *BA*
GPA: 4.0

Providence, RI
Graduated: 05/23

Work Experience

Google 12/24 - 07/26
Software Engineer | Ads Sunnyvale, CA

- Delivered an end-to-end manual-release testing automation platform from 0-to-1 in 9 months, aligning 3+ cross-functional stakeholder teams on a shared rollout timeline and authoring 4 technical design documents and proposals; eliminated a 24-hour manual bottleneck, achieved 100% test coverage, and saved \$60K annually.
- Delivered a performance-metrics data pipeline from design through launch by building a custom device-metrics crawler and parser, automating Jank, memory, and latency tracking and cutting manual triage time by 30%.
- Built an agentic test-analysis and stress-testing pipeline across 5 ad formats that autonomously analyzed logs and surfaced SDK breaking points.
- Cut the failing-test backlog by 50% (130 to 65) and processed 500+ bug reports through daily Python-based triage automation.

AIM Executive Coaching 09/24 - 10/24
Full Stack Web Developer | Paid Independent Contract Remote

- Built and launched a responsive client website with HTML, CSS, JavaScript, and HubSpot, increasing discoverability by 21% through SEO, performance optimization, and cross-device development.
- Translated client requirements and wireframes into a user-facing product, iterating on feedback to improve usability and alignment with engagement goals.

Relevant Projects

Full Stack & Agentic Software Engineering

- Designed an agentic pipeline that autonomously analyzes CI/CD build and test failures, using LLM-driven reasoning to classify root causes and surface actionable triage recommendations.
- Produced a full-stack web application using MongoDB, Node.js, Express, and Vue.js with a RESTful API architecture.

Applied Machine Learning & Predictive Modeling

- Built and tuned an XGBoost/ChemBERTa ensemble on 8,703 raw records (cleaned to 5,565), achieving a scaffold-split test RMSE of 0.82 ($R^2=0.55$, Spearman $\rho=0.72$), beating a naive baseline (RMSE 1.23), with results validated via 1,000-sample bootstrapping.
- Developed multiclass classification models with severe class imbalance and missing-data imputation, achieving a 15% improvement in prediction accuracy.

Master's Dissertation

Machine Learning for Medical Image & Data Analysis (Joint Dissertation) 01/24 - 08/24
University of St Andrews

- Co-developed computer vision (YOLOv8) and ML models over 200,000+ tabular and image data points, using generative AI for data augmentation and improving detection accuracy by 25%.
- Cut processing time by 60% by containerizing the pipeline with Docker and deploying on GPU infrastructure; maintained reproducibility via Git version control.
- Built user-facing tools exposing supervised and unsupervised models to end users, reducing analysis time by 30%.

Technical Skills

Programming: Python, Java, JavaScript, HTML, CSS, SQL, Kotlin, Swift, Go

Development: Integration Testing, Performance Testing, Release Engineering, Physical Device Testing, Large-Scale Test Infrastructure, Node.js, Express, Vue.js, React, Angular, RESTful APIs, MongoDB, MySQL, D3.js, Database Management (Relational and NoSQL), Firebase, Distributed Systems, On-Call Rotation, End-To-End Project Ownership, Kubernetes, GraphQL

Machine Learning & AI: pandas, scikit-learn, TensorFlow, PyTorch, Keras, Deep Learning, LLM, Machine Learning Pipelines, Data Pipelines, Prompt Engineering, AI Fluency, Agentic Workflows, LLM Integration, Gemini API, AI-Assisted Development, Cursor, Claude Code, Codex

Tools: Git, GitHub, Docker, Containers, Jupyter Notebook, Tableau, Batch Job Schedulers, Cloud Infrastructure, Linux

Practices & Specializations: Frontend, Backend, Full-Stack, Mobile Development, Android, iOS, Object-Oriented Programming, Agile, Scrum, Microservices, Pair Programming, User-Centric Design, CI/CD, Release Automation, DevOps, On-Call Rotation, Bazel/bzl